

# NABEEL REHEMTULLA

## CURRICULUM VITAE

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### EDUCATION

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|            |                                                                                                                                        |                      |
|------------|----------------------------------------------------------------------------------------------------------------------------------------|----------------------|
| <i>PhD</i> | Northwestern University, Astronomy<br>Presidential Fellow                                                                              | <i>Expected 2027</i> |
| <i>MS</i>  | Northwestern University, Astronomy<br>Integrated Data Science Certificate                                                              | 2023                 |
| <i>BS</i>  | University of Michigan, Astronomy & Astrophysics with Highest Honors<br>Second Major Interdisciplinary Physics; Minor Computer Science | 2021                 |

### REFEREED FIRST-AUTHOR PAPERS

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Li\*, Chen\*, **Rehemtulla\***, et al. 2026, "*StarEmbed: Benchmarking Time Series Foundation Models on Astronomical Observations of Variable Stars*," Proceedings of the 43<sup>rd</sup> International Conference on Machine Learning, arXiv:2510.06200

\*Equal contribution

**Rehemtulla**, Miller, Walmsley, et al., 2026, "*Pre-training vision models for the classification of alerts from wide-field time-domain surveys*," PASP, 138, 034503

**Rehemtulla**, Coughlin, Miller, et al. 2025b, "*The automation of optical transient discovery and classification in Rubin-era time-domain astronomy*," NatAs, 9, 1764R

**Rehemtulla**, Jacobson-Galán, Singh, et al. 2025a, "*The BTSbot-nearby Discovery of SN 2024jlf: rapid, autonomous follow-up probes interaction in an 18.5 Mpc Type IIP supernova*," ApJ, 985, 241

**Rehemtulla**, Miller, Jegou du Laz, et al., 2024, "*The Zwicky Transient Facility Bright Transient Survey. III. BTSbot: Automated Identification and Follow-up of Bright Transients with Deep Learning*," ApJ, 972, 7

**Rehemtulla**, Miller, Coughlin, et al., 2023, "*BTSbot: A Multi-input Convolutional Neural Network to Automate and Expedite Bright Transient Identification for the Zwicky Transient Facility*," Proceedings of the ML for Astrophysics Workshop at the 40<sup>th</sup> International Conference on Machine Learning, arXiv:2307.07618

**Rehemtulla**, Valluri & Vasiliev, 2022, "*Non-parametric spherical Jeans mass estimation with B-splines*," MNRAS, 511, 5536

### AWARDS AND FELLOWSHIPS

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|                                                                                                                                   |      |
|-----------------------------------------------------------------------------------------------------------------------------------|------|
| <b>Outstanding Paper Award</b><br>International Conference on Machine Learning – AI4Physics Workshop                              | 2026 |
| <b>Presidential Fellowship</b><br>Northwestern University – the <i>highest honor Northwestern bestows upon a graduate student</i> | 2025 |
| <b>Outstanding Student Achievement Award</b><br>Astroinformatics Conference, Puerto Natales, Chile                                | 2024 |

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|------------------------------------------------------------------------------------|------------|
| <b>2<sup>nd</sup> prize, Rapid Fire Research 2024</b>                              | 2024       |
| Northwestern University, Department of Physics and Astronomy                       |            |
| <b>1<sup>st</sup> prize, Visualization Competition (<a href="#">animation</a>)</b> | 2024       |
| Northwestern University, Computation and Data Exchange ( <a href="#">CoDEx</a> )   |            |
| <b>NASA Illinois Space Grant Consortium Graduate Fellowship</b>                    | 2023, 2022 |
| (awarded twice)                                                                    |            |
| <b>Highest Honors in Astronomy &amp; Astrophysics</b>                              | 2021       |
| University of Michigan, Department of Astronomy                                    |            |
| <b>Chambliss Astronomy Achievement Award</b>                                       | 2021       |
| American Astronomical Society                                                      |            |
| <b>Outstanding Undergraduate Poster</b>                                            | 2020       |
| Michigan Institute for Data Science                                                |            |

#### AWARDED TELESCOPE TIME AS PI

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|--------------------------------|----------------------|
| PI 1.6 n; SOAR Telescope       | NOIRLab   2026B      |
| PI 2.0 n; MMT Observatory      | Northwestern   2026B |
| PI 1.0 n; Gemini-N Observatory | NOIRLab   2026A      |
| PI 1.3 n; SOAR Telescope       | NOIRLab   2026A      |
| PI 0.4 n; SALT Observatory     | NOIRLab   2026A      |
| PI 2.5 n; MMT Observatory      | Northwestern   2026A |
| PI 1.0 n; Gemini-N Observatory | NOIRLab   2025B      |
| PI 3.5 n; MMT Observatory      | Northwestern   2025B |
| PI 1.0 n; SOAR Telescope       | NOIRLab   2025A      |
| PI 3.0 n; MMT Observatory      | Northwestern   2025A |

#### INVITED TALKS & SEMINARS

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|-----------------------------------------------------------------------|------------------------|
| Invited Lecture, Rise_Time 2026 – Precursors: Soiree for Rising Stars | Remote; Aug 2026       |
| Seminar, Carnegie High Energy & Massive Stars Group Meeting           | Pasadena, CA; May 2026 |
| Seminar, Caltech Special Seminar                                      | Pasadena, CA; May 2026 |
| Seminar, NOIRlab FLASH Seminar                                        | Tucson, AZ; Apr 2026   |
| Invited Talk, Earth 2.0 Telescope Science Seminar                     | Remote; Jan 2026       |
| Invited Talk, CCA Foundation Models for Astronomy                     | New York, NY; May 2025 |
| Invited Talk, LS4 Team Meeting                                        | Evanston, IL; Mar 2025 |
| Invited Talk, Rutgers University Astronomy Journal Club               | Remote; Feb 2025       |
| Invited Talk, A3D3 Multi-messenger Astrophysics Monthly Telecon       | Remote; Nov 2023       |

#### CONTRIBUTED TALKS, POSTERS, AND PANELS

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|------------------------------------------------------------------------|----------|
| Talk, Empowering Science in the Data-rich Era of Astronomy, Tucson, AZ | Apr 2026 |
| Talk, RAPID Response: Hotwiring Transients, Pasadena, CA               | Mar 2026 |
| Talk, TDABench 2026, Chicago, IL                                       | Mar 2026 |
| Talk, Astroinformatics 2026, Cologne, Germany                          | Feb 2026 |
| Poster, Open SkAI 2025, Chicago, IL                                    | Sep 2025 |

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|----------------------------------------------------------------------------|----------|
| Talk & Panelist, Rutgers Summer Transient Soiree, Piscataway, NJ           | Jul 2025 |
| Talk, Celebrating 20 Years of Swift Discoveries, Florence, Italy           | Mar 2025 |
| Talk, 245 <sup>th</sup> AAS Meeting, National Harbor, MD                   | Jan 2025 |
| Talk, Astroinformatics 2024, Puerto Natales, Chile                         | Dec 2024 |
| Poster, Rise_Time 2024, West Lafayette, IN                                 | Aug 2024 |
| Poster, International Conference on Machine Learning, Honolulu, HI         | Jul 2023 |
| Talk, The Transient and Variable Universe, Urbana, IL                      | Jun 2023 |
| Lecture, ASTRON 111 – Introduction to Astrobiology, Evanston, IL           | Apr 2023 |
| Poster, 241 <sup>st</sup> AAS Meeting, Seattle, WA                         | Jan 2023 |
| Poster, 237 <sup>th</sup> AAS Meeting <i>DESI Special Session</i> (remote) | Jan 2021 |

## PRESS

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Total reach of >3.1M across >30 media mentions

Selected articles featuring "*SN 2023tyk: discovery to spectroscopic classification performed fully automatically*" (Rehemtulla et al. 2023)

- ◆ [Northwestern Now](#), “*First supernova detected, confirmed, classified and shared by AI*”
- ◆ [Space.com](#), “*AI just spotted its 1st supernova. Could it replace human explosion hunters?*”
- ◆ [Gizmodo](#), “*AI Detected a Supernova All on Its Own*”
- ◆ [Dropbox Blogs](#), “*An AI-powered telescope is helping astronomers sift through the stars*”

Selected articles featuring "*BTSbot: Automated Identification and Follow-up of Bright Transients with Deep Learning*" (Rehemtulla et al. 2024)

- ◆ [Northwestern Now](#), “*Sky survey classifies more than 10,000 supernovae*”
- ◆ [Caltech News](#), “*ZTF Leads to Classification of 10,000 Supernovae*”

## SELECTED FIRST-AUTHOR ASTRONOTES

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AstroNotes are circulars to the transient astronomy community

**Rehemtulla**, Jacobson-Galán, Kwok, et al., 2026, "Reclassification of SN 2026sqf, the nearest Type Ia-CSM to-date," TNSAN, 2026-240

**Rehemtulla**, Sollerman, Liu, et al., 2025b, "Discovery and classification of SN 2025ahqr, a highly reddened low luminosity Type Iax in M108 (9 Mpc)," TNSAN, 2025-345

**Rehemtulla**, Schulze, Jacobson-Galán, et al., 2025a, "Swift/UVOT observations of SN 2025qtt 6 minutes after its automatic discovery by BTSbot-nearby," TNSAN, 2025-209

**Rehemtulla**, Miller, Fremling, et al., 2023, "SN 2023tyk: discovery to spectroscopic classification performed fully automatically," TNSAN, 2023-265

- ◆ 5<sup>th</sup> most viewed AstroNote of all time

## MENTORING & LEADERSHIP

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**Northwestern University, CIERA, SkAI**

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|--------------------------------------------------------------------------|-------------|
| Satellite Early Education for Discovery (SEED) REU Program, SkAI         | Summer 2026 |
| Graduate Student Council Mentorship Committee, Dept. of Physics & Astro. | 2025–2026   |
| Code/Astro Workshop, Teaching Assistant                                  | Summer 2023 |
| ASTRON 111, Teaching Assistant                                           | Spring 2023 |
| ASTRON 120, Teaching Assistant                                           | Fall 2022   |

Research Experiences at CIERA for High Schoolers, Project Mentor

Summer 2022

**Ismaili Center of Chicago (local mosque), Chicago, IL**  
Professionals Network, Midwest - Board Member

September 2023 – Present

## CO-AUTHOR PUBLICATIONS

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\*Significant contribution

Sevilla, Ho, Nayana, et al., "*Multiwavelength Analysis of Six Luminous, Fast Blue Optical Transients*," 2026, ApJ, 1007, 38S

Sasli, Xu, Junell, et al., "*Catching Disguised Transients with ASTRANet: Anomaly-Aware Spectroscopic Classification and Conformal Calibration*," 2026, arXiv:2607.08044

Simongini, Acero, Imbrogno, et al., "*Bridging the gap between SLSNe and SE-SNe. Multi-wavelength analysis of the SLSN-Ib SN 2024jlc*," 2026, arXiv:2607.01053

Kwok, Liu, Jha, et al., "*JWST Spectroscopy of SN Ia 2022aaiq and 2024gy: Evidence for Enhanced Central Stable Ni Abundance and a Deflagration-to-Detonation Transition*," 2026, ApJ, 1005, 52K

\*Shah, Rehemtulla, Miller, et al., "*Leveraging Multimodality for Real-Time Classification of Transients and Variables found by the Zwicky Transient Facility*," 2026, arXiv:2607.00228

Perley, Ho, McGrath, et al., "*AT2024wpp: An Extremely Luminous Fast Ultraviolet Transient Powered by Accretion onto a Black Hole*," 2026, MNRAS, 549, stag678

Junell, Sasli, Nunes, et al., "*Applying multimodal learning to Classify transient Detections Early (AppleCiDER) I: Data set, methods, and infrastructure*," 2026, PASP, 138, 054508

Singh, Kwok, Jha, et al., "*Photometry and Spectroscopy of SN 2024pxl: A Luminosity Link Among Type Iax Supernovae*," 2026, ApJ, 999, 227

Barjou-Delayre, Rosnet, Ravoux, et al., "*Study of the anisotropy of cosmic expansion on ZTF type Ia supernovae simulations*," 2026, arXiv:2601.11139

Burgaz, Maguire, Galbany, et al., "*ZTF SN Ia DR2 follow-up: Exploring the origin of the Type Ia supernova host galaxy step through Si II velocities*," 2026, A&A, 705, A76

Kasliwal, Ahumada, Stein, et al., "*ZTF25abjmnps (AT2025ulz) and S250818k: A Candidate Superkilonova from a Subthreshold Subsolar Gravitational-wave Trigger*," 2025, ApJ, 995, L59

Pearson, Subrayan, Sand, et al., "*Mid-infrared Dust Evolution and Late-time Circumstellar Medium Interaction in SN 2017eaw*," 2025, ApJ, 993, 213

Xu, Sasli, Junell, et al., "*AppleCiDER II: SpectraNet -- A Deep Learning Network for Spectroscopic Data*," 2025, arXiv:2510.07215

Miller, Abrams, Aldering, et al., "*The La Silla Schmidt Southern Survey*," 2025, PASP, 137, 094204

\*Gupta, Muthukrishna, Rehemtulla, et al., "*Transfer learning for transient classification: from simulations to real data and ZTF to LSST*," 2025, MNRAS, 542, L132

\*Medina, Li, Eadie, et al., "*The mass of the Milky Way from outer halo stars measured by DESI DRI*," 2025, arXiv:2508.19351

Kwok, Singh, Jha, et al., "*JWST and Ground-based Observations of the Type Ia Supernovae SN 2024pxl and SN 2024vjm: Evidence for Weak Deflagration Explosions*," 2025, ApJ, 989, L33

Yao, Chornock, Ward, et al., "*A Massive Black Hole 0.8 kpc from the Host Nucleus Revealed by the Offset Tidal Disruption Event AT2024tvd*," 2025, ApJ, 985, L48

Das, Kasliwal, Fremling, et al., "*Low-luminosity Type IIP Supernovae from the Zwicky Transient Facility Census of the Local Universe. I. Luminosity Function, Volumetric Rate*," 2025, PASP, 137, 044203

Sharma, Mahabal, Sollerman, et al., "*CCSNscore: A Multi-input Deep Learning Tool for Classification of Core-collapse Supernovae Using SED-machine Spectra*," 2025, PASP, 137, 034507

Sharma, Sollerman, Kulkarni, et al., "*Dramatic Rebrightening of the Type-changing Stripped-envelope Supernova SN 2023aew*," 2024, ApJ, 966, 199

Strotjohann, Ofek, Gal-Yam, et al., "*Search for Supernova Progenitor Stars with ZTF and LSST*," 2024, ApJ, 960, 72

Cooper, Kopolov, Allende Prieto, et al., "*Overview of the DESI Milky Way Survey*," 2023, ApJ, 947, 37